SI-C-002 — Perspective-Limited Truth and Open Boundaries of Intelligence
From Computational Perspective to Boundary Intelligence

Sizhe Tan
with chatGPT (OpenAI)
2026-08-16

Repository: https://github.com/sizhet/Structural-Intelligence-Compass

Abstract
Intelligent systems do not access the world from nowhere.
They observe through measurements, represent through models, compute through algorithms, traverse through Function Tunnels, and validate through finite evidence.
Every such process creates a computational perspective.
A perspective may preserve important structure and support highly reliable conclusions. But the success of a perspective does not, by itself, establish that the perspective is lossless or complete.
This paper develops a structural view of truth and intelligent boundaries.
It distinguishes **perspective-relative truth** from claims of absolute truth, introduces the **Perspective-Loss Principle** and **Cross-Perspective Recovery Principle**, and argues that increasingly capable intelligence should not be evaluated by whether it eliminates uncertainty, but by whether it can recognize, represent, test, communicate, and expand the validity boundaries of its own knowledge and computation.
This capability is referred to as **Boundary Intelligence**.
The paper also argues that even highly capable Hybrid AI or future AGI should be understood primarily in terms of coverage, adaptability, reliability, validation, and utility rather than as a mechanism that has acquired ownership of truth.
The central conclusion is:
A more general intelligence does not terminate the unknown. It becomes better able to organize and move the boundary between the known and the unknown.

1. Intelligence Always Sees Through Something

Any intelligent result depends on a path between an underlying problem and the answer produced.
That path may include:

- observation;
- measurement;
- sensing;
- representation;
- feature selection;
- embedding;
- model structure;
- training data;
- algorithmic assumptions;
- Function Tunnels;
- runtime state;
- validation procedures.

There is therefore no practical intelligent computation without mediation.
At a high level:
Underlying Structure
→ **Observation / Representation**
→ **Computational Perspective**
→ **Result**
This does not mean that the result is arbitrary.
It means that the result is obtained through a specific access structure.
The central question becomes:
How much of the underlying structure survives that access process?

2. Perspective Is Not the Same as Error

The word *perspective* should not be interpreted as meaning subjective, unreliable, or false.
A perspective can be extremely accurate.
A carefully constructed telescope gives a perspective.
A microscope gives a perspective.
A statistical model gives a perspective.
A causal model gives a perspective.
A Large Language Model gives a perspective.
A World Model gives a perspective.
A human expert gives a perspective.
A Function Tunnel gives a perspective.
The important distinction is not:
Perspective vs. Truth

but:
Perspective-Limited Access vs. Assumed Complete Access
A perspective can preserve enough structure to support a strong truth claim within its validated domain.
The danger begins when:
successful access is silently promoted into complete access.

3. Relative Truth and Absolute Truth

Suppose an intelligent system observes a problem through one computational tunnel.
The tunnel produces a consistent answer.
Repeated tests confirm the result.
This establishes a meaningful form of truth.
But what kind?
One possibility is:

Perspective-Relative Truth

A conclusion that is valid relative to:

- the chosen representation;
- the observed variables;
- the active Function Tunnel;
- the current structural assumptions;
- the validated domain.

Such truth may be practically strong.
It may support engineering, prediction, control, and decision making.
But another conclusion does not automatically follow:
Because this perspective works, no other perspective contains additional relevant structure.
That second claim is much stronger.

4. The Single-Tunnel Illusion

A computational system may become so capable within one family of paths that its boundaries become difficult to see.
Consider:
Input
→ **Representation**
→ **Function Tunnel**
→ **Output**
If the tunnel solves thousands of tasks, users may begin to infer:
The tunnel is becoming general.
That may be true in a useful capability sense.

But a second inference often appears:
The tunnel increasingly represents the world itself.
This is not necessarily justified.
The tunnel may simply have become extraordinarily broad.
A useful analogy is a monkey learning to retrieve bananas through an increasingly large network of reachable chains.
The monkey may become highly competent.
The accessible banana set may become enormous.
But:
A very large reachable set is still not evidence that no unreachable region exists.
The same applies computationally.

5. Projection and Loss

Every representation emphasizes some distinctions and suppresses others.
A map preserves certain relations.
It omits others.
A projection of a three-dimensional object preserves some geometry while losing depth.
A language description preserves some conceptual structure while omitting continuous sensory detail.
An embedding preserves some relational information while compressing the original space.
A learned model preserves patterns that help solve its training objective while discarding information that is not required for that objective.
This motivates:

Perspective-Loss Principle

Any computational or observational perspective may preserve some relevant structure while losing other relevant structure.
The word *may* matters.
Some observations are effectively lossless for the purpose at hand.
Others are not.
The principle does not assert universal severe loss.
It asserts that loss cannot be dismissed merely because a perspective is useful.

6. Loss Is Task-Dependent

A representation can be lossless for one task and lossy for another.
Consider a structural drawing.
For determining beam length, it may preserve everything needed.

For analyzing surface texture, it may preserve almost nothing.
Likewise, an embedding may be excellent for semantic retrieval but poor for preserving exact symbolic order.
A compressed runtime representation may be excellent for prediction but poor for reconstructing the original causal path.
Therefore:
Loss should be evaluated relative to the computational question being asked.
This leads to a useful distinction:
Information Loss
and:
Task-Relevant Structural Loss
are not identical.
Intelligence is primarily concerned with the second.

7. Measurement and the Limits of Passive Observation

Classical intuition often assumes that the observer can stand outside the system.
Under many macroscopic conditions, this is an excellent approximation.
A measuring instrument may perturb a system so little that the measured value can be treated as a direct reading of a pre-existing state.
This encourages a conceptual pattern:
Object
→ **Measurement**
→ **Observed Property**
where the measurement is treated as nearly transparent.
At other scales and under other measurement conditions, this assumption becomes less secure.
Quantum measurement provides a particularly sharp example.
The significance here is not a specific interpretation of quantum mechanics.
The structural lesson is narrower:
Observation need not always be separable from observation context.
When the interaction required to obtain information becomes part of the phenomenon being described, one should be cautious about treating the observed value as a lossless window onto a context-independent classical state.

8. What Quantum Measurement Contributes to This Discussion

Quantum mechanics should not be reduced to the statement that

"measurement causes noise."
That would be too weak.
Nor should Structural Intelligence attempt to resolve foundational disputes in physics.
The useful lesson is methodological.
Quantum theory reminds us that:

- measurement context matters;
- not all observables can be treated as simultaneously available classical properties;
- highly precise physical law does not require deterministic prediction of every individual outcome;
- probability distributions themselves can be law-governed and exact.

This supports an important correction:
Truth does not require deterministic point prediction.
A theory may be extraordinarily precise about structure, constraints, probability distributions, or transition rules while remaining unable to predict one future outcome with certainty.
This matters far beyond physics.

9. Structural Truth Can Exceed Point Prediction

Suppose an intelligent system cannot predict the exact next outcome.
It may still know:

- which outcomes are possible;
- which outcomes are impossible;
- which relationships are invariant;
- which transitions are strongly constrained;
- which distributions are stable;
- which structural conditions change the result.

This suggests that intelligence should not equate:
truth
with:
single deterministic answer.
A broader structural conception of truth may include:
invariants
constraints
causal relationships
probability structures
validity boundaries
runtime regularities
This is particularly relevant when direct point prediction is impossible or

inappropriate.

10. Multiple Perspectives

If one perspective may be lossy, a natural response is to add another perspective.
But simple accumulation is not enough.
Suppose four perspectives examine one problem:
P1
P2
P3
P4
The key question is not merely:
Which answer receives the most votes?
A structurally stronger question is:
What is preserved across these perspectives, and what changes?
This motivates:
Multiple Perspectives
→ **Preserve**
→ **Subtract**
→ **Cross-Perspective Structure**
→ **Improved Approximation**

11. Cross-Perspective Recovery Principle

This leads to a second principle:

Cross-Perspective Recovery Principle

When no single perspective can be assumed to be structurally lossless, comparison among structurally different perspectives can improve recovery of preserved and differential structure.
The purpose is not to produce a metaphysical guarantee.
The purpose is computational.
Different perspectives can reveal:

- common invariants;
- perspective-specific artifacts;
- hidden variables;
- boundary conditions;
- alternative causal explanations;
- missing information;
- unstable conclusions.

The comparison itself becomes a source of intelligence.

12. Preserve and Subtract

A particularly useful structural comparison is:

Preserve

What remains common across two perspectives?

Subtract

What appears in one but not the other?

This is closely related to Two-Way CCC.

The logic is simple:

Perspective A

vs.

Perspective B

→ **Preserved Structure**

- **Differential Structure**

The preserved structure may indicate:

- robust invariants;
- common causal mechanisms;
- shared constraints.

The differential structure may indicate:

- perspective bias;
- changed boundary condition;
- missing variable;
- trigger condition;
- representation effect.

This can transform disagreement from a problem into a signal.

13. Perspective Diversity Matters More Than Perspective Count

Ten nearly identical perspectives may reveal less than two structurally different ones.

Therefore:

Perspective diversity matters.

If all models share:

- the same data;
- the same representation;
- the same objective;
- the same architecture;
- the same hidden assumptions,

then agreement may provide less independent evidence than it appears to.

This is especially important in AI ensembles.

Multiple models are not necessarily multiple perspectives in a deep structural

sense.
A stronger system may need diversity in:

- representation;
- computational path;
- measurement;
- causal assumptions;
- data source;
- runtime organization.

14. Truth Approximation Rather Than Truth Ownership

The structural goal of multiple perspectives can therefore be expressed as:
More Diverse Perspectives
→ **More Cross-Checks**
→ **More Preserved Structure**
→ **More Explicit Differences**
→ **Better Structural Approximation**
But not:
Enough Perspectives
→ **Guaranteed Absolute Truth**
This distinction is central.
An intelligence system can improve continuously without reaching an epistemically closed state.
That is not a weakness.
It may be the normal condition of finite intelligence.

15. AGI Does Not Require Truth Ownership

Discussions of Artificial General Intelligence sometimes blur two different ideas:

General Capability

The ability to perform broadly across many tasks.
and:

Epistemic Completeness

The ability to possess or reconstruct complete truth.
The first is meaningful as an engineering and behavioral objective.
The second is far stronger.
There is no reason to assume that achieving the first automatically produces the second.
A highly capable AGI may:

- solve many tasks;
- adapt to unfamiliar situations;

- use tools;
- build models;
- plan;
- reason;
- communicate;
- learn;
- revise itself;
- coordinate with other systems.

None of this logically establishes omniscience.

16. AGI as a Capability Concept

A more useful conception of AGI may focus on dimensions such as:

- task coverage;
- adaptability;
- learning efficiency;
- transfer capability;
- reliability;
- robustness;
- tool use;
- autonomous organization;
- collaborative ability;
- uncertainty recognition;
- boundary recognition;
- capacity to acquire new perspectives.

These can be measured relative to human needs and practical use.
In this sense:
AGI is better treated as a general-capability concept than as a certificate of access to absolute truth.
Its evaluation is fundamentally functional.

17. Human Convenience and General Intelligence

General intelligence is meaningful partly because humans care about what the system can do for them.
We ask whether a system can:

- help solve scientific problems;
- write software;
- diagnose failures;
- organize information;
- operate tools;
- assist decisions;

- adapt to new tasks;
- communicate across domains.

These are utility and coverage questions.
They are important.
But they are not equivalent to:
Does the system possess the final truth about the world?
A system may become dramatically more useful without moving proportionally closer to epistemic completion.

18. The Boundary Must Remain Open

Every finite intelligent system operates within some boundary.
That boundary may reflect:

- limited observation;
- limited representation;
- limited data;
- limited compute;
- limited models;
- limited validation;
- unobserved future conditions;
- unknown phenomena.

A common mistake is to treat the boundary as a temporary defect to be eliminated.
A more mature view is:
The boundary is itself part of the state of intelligence.
The system should know where its validated knowledge ends.

19. Boundary Intelligence

This motivates a distinct concept.

Boundary Intelligence

Boundary Intelligence is the capability of an intelligent system to recognize, represent, test, communicate, and expand the validity boundaries of its own knowledge and computation.
Boundary Intelligence includes questions such as:

- What do I know?
- Why do I know it?
- From which perspective?
- Under which conditions?
- How far has it been validated?
- What assumptions are required?
- What changes would invalidate it?

- Which alternative perspectives disagree?
- Where am I extrapolating?
- What remains unknown?

This is not an optional philosophical layer.
It can directly affect runtime behavior.

20. Unknown as a Legitimate Runtime State

An intelligent system should not always be forced into one of two states:
answer
or:
failure.
A more mature runtime may include:

- known;
- probable;
- structurally supported;
- weakly supported;
- disputed;
- out-of-domain;
- unresolved;
- unknown.

The state:
Unknown
can itself activate computation.
For example:
Unknown
→ **Request Another Perspective**
→ **Run Another Function Tunnel**
→ **Call World Model**
→ **Use Tool**
→ **Acquire Data**
→ **Run Simulation**
→ **Ask Human**
→ **Validate**
Unknown becomes a trigger.

21. Boundary as Trigger

This leads to an important Structural Intelligence idea:
The detection of a boundary can trigger a change in computational organization.
Suppose a local intelligence node detects:

Low Structural Confidence
The runtime system may:
- switch decision engines;
- expand search;
- activate another node;
- invoke another model;
- escalate to a human;
- collect more evidence.

Boundary awareness therefore connects naturally to:
- Calling Graphs;
- FTRI switching;
- PDOS routing;
- Local Node Intelligence;
- Hybrid AI.

Epistemology becomes runtime architecture.

22. Open Boundary and Future Knowledge
A boundary is not only the edge of current knowledge.
It is the interface through which future knowledge enters.
New evidence may change:
- classification;
- causal interpretation;
- structural confidence;
- runtime policy;
- model organization.

Therefore:
The boundary must remain shareable with the future.
This phrase has a precise meaning.
A mature intelligence system should be designed so that:
- new evidence can revise existing conclusions;
- new perspectives can be added;
- new structures can be named;
- new local intelligence mechanisms can be integrated;
- old assumptions can be invalidated;
- unknown states can become known.

A closed boundary blocks learning.

23. The Future Is Not Just More Data
The future may contribute more than additional examples.
It may contribute:

- new measurement methods;
- new representations;
- new causal variables;
- new computational primitives;
- new theories;
- new forms of intelligence;
- new categories not present in the original model.

This is especially important.
If a system only expects:
more of the same data
it may remain trapped within its current perspective.
Open intelligence must allow:
new kinds of structure to enter.

24. World Models Reconsidered

The idea of an open boundary also affects how World Models may be understood.
A naive interpretation imagines a World Model as:
an internal complete copy of the world.
A more realistic structural interpretation is:
a dynamically organized collection of validated perspectives, invariants, relations, transitions, expectations, and boundaries about the world.
Such a World Model may contain:

- structural relations;
- causal relations;
- probability distributions;
- local models;
- alternative perspectives;
- uncertainty;
- validity domains;
- runtime expectations.

Its strength would come not from claiming totality, but from managing structured incompleteness well.

25. Hybrid Intelligence as Perspective Organization

Hybrid Intelligence becomes especially relevant under this view.
A hybrid system may combine:

- LLM;
- World Model;
- symbolic reasoning;

- causal models;
- simulators;
- structural models;
- tools;
- humans.

The benefit is not merely that there are more components.
The deeper opportunity is:
Different intelligence mechanisms can supply structurally different perspectives.
A runtime organization can compare them.
This enables:
Perspective Diversity
→ **Cross-Perspective Validation**
→ **Boundary Detection**
→ **Improved Decision**

26. But Hybrid Intelligence Still Does Not Own Truth

Suppose a Hybrid AI system becomes extremely capable.
It has:

- many models;
- many tools;
- many sensors;
- many perspectives;
- strong validation;
- extensive memory;
- adaptive runtime organization.

Its epistemic coverage may become enormous.
But finite hybridization does not logically imply complete coverage.
Therefore:
Hybrid Intelligence can reduce perspective loss without eliminating the concept of perspective.
This is why Point 12 of the Structural Intelligence Compass returns to Point 1.

27. Boundary Expansion Is Not Boundary Elimination

A mature intelligence system should attempt to expand its valid domain.
That process may look like:
Known Region
→ **Boundary Detection**
→ **New Observation**
→ **Alternative Perspective**

→ **Structural Comparison**
→ **Validation**
→ **Expanded Known Region**
But a new boundary appears.
Therefore:
Boundary Expansion
does not imply:
Boundary Elimination.
This process can continue indefinitely.

28. Intelligence as Boundary Reorganization

This leads to a broader definition.
Intelligence can be viewed not merely as answer production, but as the repeated reorganization of the boundary between:
Known
Uncertain
Unknown
The cycle becomes:
Observe
→ **Represent**
→ **Organize**
→ **Infer**
→ **Act**
→ **Measure**
→ **Validate**
→ **Recognize Boundary**
→ **Acquire New Perspective**
→ **Reorganize**
This is an open runtime loop.

29. Open Boundaries and Scientific Practice

Scientific progress already operates in a similar way.
A theory explains a domain.
New evidence reveals an anomaly.
The anomaly identifies a boundary.
A new model is proposed.
The boundary moves.
The new theory may include the previous one as a special case.
Science therefore does not progress primarily by proving that boundaries no longer exist.

It progresses by making them more explicit and moving them.
This makes Boundary Intelligence highly compatible with scientific reasoning.

30. Open Boundaries and Collective Learning

No individual intelligence is required to discover every perspective.
Collective learning can distribute perspective discovery across:

- people;
- research groups;
- disciplines;
- tools;
- AI systems;
- experiments;
- future generations.

This gives open publication a structural role.
Publishing an incomplete but well-specified boundary allows others to work beyond it.
The purpose is not merely communication.
It is **distributed boundary expansion**.

31. Perspective-Limited Intelligence Is Not Weak Intelligence

There is a temptation to think:
If intelligence has boundaries, then it is fundamentally inadequate.
This does not follow.
Human intelligence is perspective-limited.
Science is perspective-limited.
Engineering models are perspective-limited.
Yet these systems can be extraordinarily powerful.
The objective is not to remove every limitation.
It is to manage limitations intelligently.
This includes:

- knowing when a model is valid;
- knowing when to switch models;
- knowing when more data are required;
- knowing when uncertainty is irreducible;
- knowing when another perspective is valuable.

These are high-order intelligence capabilities.

32. The Danger of Perspective Closure

A particularly dangerous system is not necessarily one with a narrow

perspective.
It may be one that:
has a broad perspective but cannot recognize that it is still a perspective.
Such a system may become overconfident precisely because its coverage is impressive.
Perspective closure occurs when:
Current Model
is mistaken for:
Complete Reality
This can happen in:
- AI;
- scientific theory;
- economic models;
- political models;
- organizational systems;
- individual reasoning.

Boundary Intelligence is a defense against perspective closure.

33. Structural Confidence and Boundary Intelligence

Boundary Intelligence requires more than uncertainty probabilities.
A system should distinguish why it is uncertain.
For example:

Metric Uncertainty

The case is far from previous data.

Structural Uncertainty

Relevant invariants may not hold.

Causal Uncertainty

The mechanism may have changed.

Perspective Uncertainty

Different models disagree.

Validation Uncertainty

The decision has not been tested in comparable runtime conditions.
These forms of uncertainty imply different responses.
A single scalar confidence score may hide important structure.

34. Multi-Axis Confidence

A future Structural Intelligence system may represent confidence as a vector or structured object:

Metric Confidence
Statistical Confidence
Structural Confidence
Causal Confidence
Perspective Agreement
Runtime Validation Confidence
Boundary Distance
This would allow statements such as:
The case is statistically unfamiliar, but the structural mechanism is strongly preserved.
or:
The case is metrically familiar, but one critical boundary condition has changed.
This is a more informative form of intelligence than a single probability.

35. Truth, Utility, and Reliability

Truth, utility, and reliability should not be collapsed.
A system may be:

- useful but poorly understood;
- reliable within a narrow domain;
- structurally insightful but operationally expensive;
- statistically accurate without causal understanding;
- causally strong but data-poor.

A mature AGI evaluation should therefore remain multidimensional.
Capability should not be used as a proxy for epistemic completeness.

36. The Open-World Requirement

A truly general intelligence should be able to operate in an open world.
An open world contains:

- new objects;
- new tasks;
- new relationships;
- new concepts;
- new tools;
- new agents;
- new constraints;
- new failure modes.

Therefore a general intelligence architecture should not assume that:
all future categories are already latent inside the current model.
That may sometimes be approximately true.

It should not be a constitutional requirement.

37. Structural Accommodation of the Unknown

An open system therefore needs mechanisms for:

- new typing;
- new naming;
- new organizational nodes;
- new Function Tunnels;
- new causal structures;
- new runtime policies;
- new validation criteria.

Structures such as UTN, differential organization, Calling Graphs, and localized intelligence may help make this possible.
The objective is:
Add new structure without rebuilding the entire intelligence system.
This is one way in which Structural Intelligence may support open-world learning.

38. Open Architecture and Open Epistemology

Two kinds of openness should be distinguished.

Architectural Openness

Can new computational mechanisms be added?

Epistemic Openness

Can new information fundamentally change what the system believes and how it organizes knowledge?
A system may be architecturally modular but epistemically closed.
It may accept a new tool while forcing all results into an old worldview.
True Boundary Intelligence requires both.

39. The Boundary Should Be Explicit

Many systems have boundaries but do not represent them.
The boundary exists only implicitly as failure.
A better system makes the boundary explicit.
For example:
Validated Region A
Weakly Validated Region B
Conflicting Region C
Unknown Region D
This converts invisible failure risk into addressable structure.
Explicit boundaries can support:

- routing;
- escalation;
- additional validation;
- risk management;
- research prioritization.

40. Boundary Maps Are Temporary

An explicit boundary map should itself be treated as provisional.
The system may discover:
- the boundary was too conservative;
- the boundary was too broad;
- multiple boundaries should exist;
- one variable changes the boundary;
- one new perspective removes an apparent contradiction.

Thus:
Boundary representation is itself subject to boundary revision.
This recursive property is important.

41. Boundary Intelligence and Self-Correction

Self-correction requires more than detecting a wrong answer.
A system may need to recognize:
the structure that produced the answer is no longer valid.
This can trigger:
- model replacement;
- policy change;
- retraining;
- new data collection;
- alternative routing;
- structural decomposition.

Boundary Intelligence therefore supports deeper correction than output patching alone.

42. Open Boundaries and Digital Brain Architecture

A Digital Brain built around Structural Intelligence could treat boundaries as distributed runtime information.
Local nodes may retain:
- local validity domain;
- X–Y–M history;
- local confidence;
- known failure modes;

- triggering conditions;
- escalation routes.

Global organization can then ask:
Which node is qualified to reason about this state?
This is different from assuming one universal intelligence unit should answer every question.
Boundary knowledge becomes part of addressing.

43. Boundary-Aware Routing

A runtime system can use boundaries directly.
For example:
Input X
→ **Route to Node A**
→ Node A detects boundary violation
→ **Trigger Node B**
→ Node B provides alternative perspective
→ **Cross-Perspective Comparison**
→ **Decision or Unknown**
This is a structurally richer form of inference.
The system does not merely predict.
It organizes who is authorized to predict.

44. A Mature Intelligence Can Say "I Do Not Know"

The phrase "I do not know" can represent many different internal states.
A mature system should ideally distinguish:
I have insufficient data.
I have conflicting perspectives.
I am outside the validated domain.
The causal structure may have changed.
The problem is computationally unresolved.
The answer may be inherently probabilistic.
These are different kinds of unknown.
Representing that difference is intelligence.

45. No Final Boundary Certificate

No matter how sophisticated the system becomes, there is no obvious general procedure that can certify:
No future observation will reveal another relevant perspective.
Therefore absolute boundary closure should be treated with caution.
This does not prohibit strong claims.

It constrains their scope.
A system may legitimately say:
Within this model, domain, evidence, and validation history, confidence is extremely high.
That is stronger and more useful than pretending the scope does not exist.

46. A Structural Form of Intellectual Humility

Boundary Intelligence is sometimes described informally as humility.
But the concept here is more technical.
It is not a personality trait.
It is a computational capability.
The system represents:

- scope;
- support;
- perspective;
- uncertainty;
- invalidation conditions;
- escalation options.

This is **structured epistemic discipline**.

47. Perspective-Limited Truth as an Engineering Principle

The concept has direct engineering consequences.
A safety-critical system should not only return:
prediction = Y
It should also retain:
source perspective
structural assumptions
validation boundary
confidence dimensions
fallback path
trigger condition
This transforms truth management into system architecture.

48. The Open-Boundary Principle

The paper can now state a broader principle:

Open-Boundary Principle

Any finite intelligent system should be treated as operating within revisable epistemic and computational boundaries, and its architecture should allow those boundaries to be recognized, challenged, and expanded.

This principle is compatible with increasingly capable AI.
Indeed, capability may depend on it.

49. The Boundary-Sharing Principle

A further extension is:

Boundary-Sharing Principle

A mature intelligence should preserve enough explicit information about its current validity boundaries that future evidence, future tools, future researchers, or future intelligence systems can revise and extend them.
This makes the future an active participant in present architecture.
The system should not merely store conclusions.
It should preserve the structure of where those conclusions may fail.

50. From Truth Ownership to Truth Participation

This leads to a different ideal for intelligence.
Instead of:
Own the truth
the objective becomes:
Participate in an expanding process of truth approximation, structural validation, and boundary revision.
This is compatible with:

- individual intelligence;
- AGI;
- scientific communities;
- collective learning;
- long-lived Digital Brain systems.

It is also more robust to future discovery.

51. The Structural Intelligence Position

The position developed here can be summarized as follows.

1. Intelligence always operates through some perspective.
2. A perspective may preserve important truth without being complete.
3. Some perspectives are lossy relative to some tasks.
4. Multiple structurally different perspectives can reveal preserved and differential structure.
5. Cross-perspective comparison can improve structural approximation.
6. Better approximation does not establish absolute truth ownership.
7. AGI should be evaluated primarily as a general capability system.
8. Mature intelligence should represent its own validity boundaries.
9. Unknown should be an explicit runtime state.

10. Boundaries should trigger new computation when appropriate.
11. Future evidence must remain able to revise current knowledge.
12. Intelligence is therefore an open process of boundary reorganization.

52. Closing Perspective

A sufficiently powerful intelligence may know more than any individual human.
It may integrate more data.
It may reason across more domains.
It may combine more perspectives.
It may validate more decisions.
It may become dramatically more reliable.
All of these developments matter.
But they should not be confused with the end of epistemic openness.
The world may continue to produce:

- new observations;
- new structures;
- new mechanisms;
- new categories;
- new tools;
- new forms of intelligence.

A mature intelligence should be designed for that possibility.
Its objective should not be to eliminate the future's ability to surprise it.
Its objective should be to remain capable of learning when surprise occurs.
More capable intelligence expands the known.
Boundary Intelligence preserves the ability to recognize what remains unknown.
And therefore:
Intelligence does not terminate the unknown. It reorganizes the boundary between the known and the unknown.

Final Principle

A more general intelligence does not own the future. It becomes better prepared to share its boundaries with the future.

Structural Intelligence Compass
SI-C-002
Perspective → Loss → Recovery → Boundary → Revision → Future